

1996, AOL systems went down for 19 hours. It resulted in loss of revenue to 16 million customers, and made a huge dent in AOL's much-publicised claims of offering the highest reliability. When the dust finally settled, it was found that the outage was not due to a hardware or software fault. It was a software upgrade failure that caused the front-page headlines, 'America Offline' (<http://catless.ncl.ac.uk/Risks/18.30.html>).

Even after rolling back the offending software upgrade, AOL could not restore its services, because the routing tables got corrupted during the upgrade process, and could not be restored immediately, resulting in a much longer outage. This is a typical example of a software upgrade scenario where the software maintainers (AOL, in this case), had not made adequate provision for an immediate roll-back in case any failures occurred during the upgrade process itself.

Another well-publicised software upgrade failure, with much greater damage to business, is the case of AT&T Wireless, well documented in the article titled 'AT & T Wireless Self-destructs', available at <http://bit.ly/kkCwS9>. This chronicles the case of a Customer Relationship Management (CRM) software upgrade that caused the company's CRM system to fail for more than two months, resulting in the loss of thousands of customers. The root cause for the outage was the breakage of dependencies between legacy back-end components, due to the upgraded software. AT & T could not roll back to the previous version, since not enough of the old version was preserved. All these issues led to the company losing \$100 million in revenue, and contributed to its ultimately getting sold off to Cingular for less than half its worth.

As we see from the above examples, in terms of their cost to the company, upgrade failures are extremely expensive and disruptive.

The main question for us to ponder over is: What are the common causes of software upgrade failures, and how can they be avoided?

Causes of software upgrade failures

Let's classify the various software upgrade faults into the following four categories:

- **Simple configuration errors:** For instance, if a start-up script has a typo, which prevents the upgrade from starting a service.
- **Semantic configuration errors:** If, for example, the maintainer performing the upgrade fails to create the required SSL certificate needed for the upgrade process to complete successfully, leading to failure.
- **Broken environmental dependencies:** Imagine a scenario in which an upgraded software requires a new library version, while a co-existing software component requires an older version. The upgrade process ends up removing the older library version and replacing it with the newer library version, causing the co-existing software component to fail because of the upgrade.
- **Data-access errors** due to software upgrades, which make the persistent data partially or fully unavailable to the application/end-user. For instance, if the upgraded application assumes a database schema that is different from (does not match) the existing database schema, this can lead to loss of data availability for the end-user.

Each of the above classes of errors can be detected, and prevented from happening, using automated tools along with greater vigilance by the software maintainer/administrator performing the software upgrade. For instance, applications should be written robustly, so that configuration errors arising from typographical mistakes or simple syntax errors can be caught and corrected. Semantic

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